Object Provenance for the Orbit-Space Gap Chain

A common page, reproduced verbatim in ai.viXra:2602.0033, 2602.0035 and 2602.0036

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Why this page exists. Four papers of this series state, prove, or consume spectral claims about objects that are routinely called "the gap on the orbit space". They are not the same object. A pointwise curvature *claim* stated in one of them was consumed, in three places, as if it were established and as if it applied to the same object. It is neither: its printed proof is not presently established, and the sound synthetic result available elsewhere concerns a measure and an operator that *have not been identified* with the ones the consuming steps need. The error was invisible because every paper used the same word. This page fixes the dictionary. **A citation discharges a hypothesis only when all four columns agree.**

Notation barrier

This page writes ν for $\nu_{\text{push}} := \pi_\# \text{Vol}_{\mathcal{A}}$, and never for anything else. The host papers of this chain use the same letter for a different object: `2602.0033` (Theorem 3.1, Step 3) and `2602.0035` (Theorem 5.3, Step 2) both write "the uniform measure ν" for the normalised Riemannian volume

$$\nu_{\text{Vol}} := d\text{Vol}_B / \text{Vol}_B(B).$$

ν_{push} **and** ν_{Vol} **are not identified**, and identifying them is exactly what (I2b) below forbids without a statement about the orbit-volume density. A reader who carries the host's ν into this page, or this page's ν into the host's Step 3, will read a printed derivation as though it already used the corrected measure. *Where the distinction bites, the subscripts are written out.*

The dictionary

Source	**Space, measure**	**Operator**	**What it can conclude**
`2602.0036`	B_{reg}, $d\text{Vol}_B$	$\mathcal{K}_B^{\text{reg}}$: unweighted Laplace–Beltrami *expression*; global realisation unspecified	*targets* a pointwise Ricci bound on the regular stratum; printed proof not presently established
`2602.0046`	B, $\nu = \pi_\# \text{Vol}_{\mathcal{A}}$	$K_\nu \geq 0$, operator of the form on $L^2(B, \nu)$	global Poincaré/LSI, singular locus included
`2602.0033`, as printed	spatial orbit space; printed *regular-stratum formula* $d\mu_{\text{print}} = Z_{\text{print}}^{-1} e^{-S_{\text{eff}}} d\text{Vol}_B$ on B_{reg}, **against** Vol_B**, not** ν; *no global extension specified*	weighted diffusion of μ_{print}, *if* a global closed form is constructed	neither a Poincaré/LSI statement nor a Hamiltonian gap is presently established
route (A): keep μ_{print}	(B, ν); a candidate *global extension* of μ_{print}, exponent $S_{\text{eff}} + \log w$	weighted form over K_ν	needs w globally, and $\text{osc}(S_{\text{eff}} + \log w)$
route (B): new measure	(B, ν); a *separately defined candidate* measure $d\widetilde{\mu} = \widetilde{Z}^{-1} e^{-S_{\text{eff}}} d\nu$	weighted form over K_ν	needs a theorem that this is the right object
`2602.0035`, as printed	$L^2(B, d\text{Vol}_B)$, global realisation not fixed; $\mu_{0,\text{print}} = \psi_{0,\text{print}}^2 \, d\text{Vol}_B$	Hamiltonian on $L^2(B, d\text{Vol}_B)$; diffusion of $\mu_{0,\text{print}}$	parts (i) and (ii) are not established by the printed proofs
candidate physical restatement	$L^2(\mathcal{A})^{\mathcal{G}} \cong L^2(B, \nu)$; $\mu_{0,\nu} = \psi_{0,\nu}^2 \, \nu$	$H_\nu = \frac{g^2}{2} K_\nu + V_{\text{pot}}$; then diffusion of $\mu_{0,\nu}$	qualitative fixed-volume positivity; a rate only under $(\text{A-osc})_\nu$

The four identifications that must never be silent

(I1) $L^2(B,\nu) \cong L^2(\mathcal{A}, \mathrm{Vol}_{\mathcal{A}})^{\mathcal{G}}$ **— exact, and free.** Pullback $f \mapsto f \circ \pi$ gives $\int_B |f|^2\,d\nu = \int_{\mathcal{A}} |f\circ\pi|^2\,d\mathrm{Vol}_{\mathcal{A}}$, which is the definition of $\nu = \pi_{\#}\mathrm{Vol}_{\mathcal{A}}$. This is an isometry, not an approximation, and it holds across the singular locus. It is the reason the metric-measure quotient is the natural physical object: gauge-invariant functions on the smooth compact $\mathcal{A}$ *are* $L^2(B,\nu)$.

(I2) K_ν **versus** $\mathcal{K}_B^{\mathrm{reg}}$ **— not free, and not even the same kind of object.** Fix the convention here, so that nothing depends on the host paper's. On the regular stratum write $d\nu = w\,d\mathrm{Vol}_B$, with w the orbit-volume density, and define the *formal differential expressions*

$$\mathcal{L}_B^{\mathrm{reg}} := \mathrm{div}_B\,\nabla, \qquad \mathcal{L}_\nu^{\mathrm{reg}} := w^{-1}\,\mathrm{div}_B(w\nabla), \qquad \mathcal{K}_\bullet^{\mathrm{reg}} := -\mathcal{L}_\bullet^{\mathrm{reg}},$$

so that

$$\mathcal{L}_\nu^{\mathrm{reg}} f = \mathcal{L}_B^{\mathrm{reg}} f + \langle \nabla \log w, \nabla f\rangle.$$

Let $K_\nu \geq 0$ denote the *global* self-adjoint operator associated with the closed Dirichlet form on $L^2(B,\nu)$.

No global self-adjoint realisation of $\mathcal{K}_B^{\mathrm{reg}}$ **is fixed by this page.** B_{reg} is the regular stratum with the singular locus removed and is in general an incomplete manifold; $\mathcal{K}_B^{\mathrm{reg}}$ on $C_c^\infty(B_{\mathrm{reg}})$ is symmetric and non-negative, but it has not been shown to be essentially self-adjoint, and it may admit several self-adjoint extensions. Choosing the Friedrichs extension is already a decision about behaviour across $B \setminus B_{\mathrm{reg}}$, and must be specified and justified separately — this is one of the gaps that forces the withdrawal of the global corollary in `2602.0036`.

The two *regular-stratum differential expressions* coincide only if w is constant on each relevant component. *Constant stabiliser type does not by itself imply constant orbit volume* — on the principal stratum all stabilisers are already conjugate, and the induced metric on the orbits, hence their volume, may still vary with the configuration; reducible connections make a global identification less tenable still. **A spectral bound for the global** K_ν **does not by itself transfer to any chosen self-adjoint realisation of** $\mathcal{K}_B^{\mathrm{reg}}$**, or conversely.** Comparison theorems under further hypotheses are not excluded; what does not exist is a free identification — and on the $\mathcal{K}_B^{\mathrm{reg}}$ side, there is not yet a single global operator for such a theorem to be about.

(I2b) $d\mathrm{Vol}_B$ **versus** ν ***at the level of the perturbed measure*** **— also not free.** This is a separate step from (I2), and it is the one most easily skipped, because both measures are written with the same exponent. On B_{reg}, $d\nu = w\,d\mathrm{Vol}_B$ with w the orbit-volume density, so the measure a paper prints against Vol_B is, written against ν,

$$d\mu_{\mathrm{print}} = Z_{\mathrm{print}}^{-1} e^{-S_{\mathrm{eff}}}\,d\mathrm{Vol}_B = Z_{\mathrm{print}}^{-1} e^{-(S_{\mathrm{eff}}+\log w)}\,d\nu \qquad \text{on } B_{\mathrm{reg}},$$

an *exact* rewriting of one and the same measure, with the same normaliser. The candidate ν-based measure $d\widetilde{\mu} = \widetilde{Z}^{-1} e^{-S_{\mathrm{eff}}} d\nu$ carries the exponent S_{eff} and, in general, a *different* normaliser. **On** B_{reg} **the printed measure has exponent** $S_{\mathrm{eff}} + \log w$ **relative to** ν**, whereas the candidate** ν**-based measure has exponent** S_{eff}**. They coincide only under an additional constancy or identification statement for** w**, none of which is established here. Replacing** $d\mathrm{Vol}_B$ **by** $d\nu$ **under a fixed exponent is therefore not a free rewriting.**

And the converse is not asserted either. This page does not claim the two measures are *proved* distinct: that would need w non-constant, up to normalisation, on the relevant domain. Reducible connections make a gratuitous global identification implausible, but implausible is not proved, and the discipline this page enforces cuts both ways — "no identification has been proved" must not become "an inequality has been proved".

Two routes are open, and they are not the same route. **(A) Keep the printed measure.** Put $\Phi_{\mathrm{print}} = S_{\mathrm{eff}} + \log w$, so that $d\mu_{\mathrm{print}} = Z_{\mathrm{print}}^{-1} e^{-\Phi_{\mathrm{print}}} d\nu$. This needs a global construction of the measure from the regular-stratum formula, control of w approaching the singular locus, and a bound on $\operatorname{osc}(S_{\text{eff}}+\log w)$ — for which a bound on $\operatorname{osc}(S_{\text{eff}})$ alone does not suffice: if w degenerates at strata of enlarged stabiliser, $\log w$ is exactly the term that destroys a bounded-perturbation argument. **No such control is supplied anywhere in this chain. (B) Change to the ν-natural measure $\widetilde{\mu}$,** which is *a separately defined candidate measure* and not the one the paper printed. Here `2602.0046` can supply the base functional inequality for K_ν, but a new theorem is required to show that $\widetilde{\mu}$, and not μ_{print}, is the correct Euclidean or transfer-theoretic object. Note that route (A) is not free either: "keep the printed measure" still means *constructing a global extension of a regular-stratum formula*, which is itself one of the open obligations. **A ν-based base inequality is therefore an input to one of these two routes, not a repair of a printed Vol_B-based derivation by change of reference measure.** Neither route by itself supplies the four-dimensional-action to three-dimensional-space step, the identification with μ_0, or a transfer-operator construction.

(I3) μ_{print} has not been identified with $\mu_0 = \psi_0^2\nu$ — not free, and the load-bearing gap. With $K_\nu \geq 0$ as in (I2) — equivalently, by (I1), the operator of the kinetic form $\mathcal{E}_\nu(f,f) = \int_B |\nabla f|^2\, d\nu$ on $L^2(\mathcal{A}, \mathrm{Vol}_\mathcal{A})^\mathcal{G}$ — **let V_{mult} be a specified real multiplication potential** and set $H_{\text{mult}} = \frac{g^2}{2}K_\nu + V_{\text{mult}}$, with a normalised, strictly positive ground state ψ_0, and put $d\mu_0 = \psi_0^2\, d\nu$. *The subscripted name is deliberate. This page is reproduced verbatim inside papers that have already bound V to a gauge-fixed potential and W to the Weyl group, so no bare letter is safe here: a page written to prevent symbol collisions must not cause one.* The ground-state transformation *then* gives

$$E_1 - E_0 = \tfrac{g^2}{2}\,\lambda_1(\mu_0), \qquad f_0 = -\log\psi_0^2,$$

where E_1 denotes the first spectral value strictly above the simple ground-state energy E_0, and, *defined explicitly to avoid the reciprocal ambiguity in the term "Poincaré constant"*,

$$\lambda_1(\mu_0) := \inf\Big\{\mathcal{E}_{\mu_0}(f,f) \;:\; f \in \operatorname{Dom}\mathcal{E}_{\mu_0},\ \textstyle\int f\, d\mu_0 = 0,\ \|f\|_{L^2(\mu_0)} = 1\Big\}.$$

That is, $\lambda_1(\mu_0)$ is the Poincaré *spectral gap* — the largest $\rho \geq 0$ with $\mathrm{Var}_{\mu_0}(f) \leq \rho^{-1}\mathcal{E}_{\mu_0}(f,f)$ for all admissible f — and *not* the constant $C_P = \rho^{-1}$ of the variance inequality, which some authors also call the Poincaré constant.

The Gibbs measure of a Euclidean effective action is *a priori* a distinct mathematical object, and its weighted Dirichlet-form gap is *a priori* a distinct spectral quantity. In the symbols fixed above,

No globally defined printed or corrected Euclidean measure has been identified with $\mu_0 = \psi_0^2\,\nu$.

That covers μ_{print}, route (A)'s global extension of it, and route (B)'s $\widetilde{\mu}$ alike — stated this way so that (I3) neither singles out one candidate nor silently presupposes (I2b). No such identification has been proved here; in special models the objects may coincide, but passing from one to the other in general requires a transfer-operator or Osterwalder–Seiler identification theorem, not a change of notation. **An oscillation bound for S_{eff} does not by itself bound $\operatorname{osc}(-\log\psi_0^2)$.**

Constants travel with their metric normalisation

$\mathrm{Ric} \geq N_c/2$ for $\langle X,Y\rangle = -\operatorname{tr}(XY)$ and $\mathrm{Ric} \geq N_c/4$ for $-2\operatorname{tr}(XY)$ are the *same* bound in two normalisations, not a sharpening of one by the other. Any statement quoting $N_c/2$ or $N_c/4$ must name its inner product.

How to use this page

Before writing that a companion result discharges a hypothesis, fill the four columns for the result and for the hypothesis. If any column differs, the citation does not discharge; at most it supplies an input to a further identification, which must then be stated as its own step. In particular:

- `2602.0046` supplies global geometry on the metric-measure object identified by (I1): a bound for K_ν on $L^2(B,\nu)$. It does not supply (I1) — that is free — and it does *not* supply a bound for any realisation of the unweighted $\mathcal{K}_B^{\mathrm{reg}}$; nor does it attempt pointwise bounds on B_{reg}, as its own text says. Note that it reaches its conclusion by synthetic quotient stability and *explicitly bypasses the O'Neill computation.* That is now the only sound geometric input in this chain: the O'Neill trace printed in `2602.0036` omits the mixed horizontal–vertical sectional terms and does not establish its own conclusion (see that record's erratum). **No pointwise Ricci bound on B_{reg} may be cited as available.**

- `2602.0033`, as printed, targets a weighted diffusion for $d\mu_{\mathrm{print}} = Z_{\mathrm{print}}^{-1} e^{-S_{\mathrm{eff}}} d\mathrm{Vol}_B$ on B_{reg} — against Vol_B, *not* against ν, and with no global extension or associated closed Dirichlet form specified. It does not presently establish a Euclidean Poincaré/LSI statement, and it does not establish a Hamiltonian or transfer-operator gap without (I3). **Citing `2602.0046` does not repair it by writing the same exponent against** ν: by (I2b) that is not a free rewriting. Either route (A) is taken and $\mathrm{osc}(S_{\mathrm{eff}} + \log w)$ is controlled, or route (B) is taken and the new measure is shown to be the right object.

- `2602.0035` v2 **already uses a ground-state measure**, so **(I3) is not the missing bridge in its printed quantitative proof** — that was the correction its v2 made to its own v1. Its printed-to-physical transition **crosses (I2) and (I2b)**: the unweighted realisation against K_ν, and $d\mathrm{Vol}_B$ against ν. The candidate physical restatement then requires $(\text{A-osc})_\nu$, a hypothesis about *its own* ground state. **Identification (I3) re-enters only if that Hamiltonian construction is used to identify a Euclidean or transfer-operator object**, as in the route towards (H1) of `2602.0033`. Among the four identifications on this page, its qualitative fixed-volume positivity needs only (I1); it additionally uses the standard compact-elliptic and positivity-improving inputs — self-adjoint realisation, compact resolvent, connectedness — and it does not require any treatment of the quotient singularities. *An earlier version of this page said flatly that the paper needs (I3) for its quantitative bound. That was wrong, and it is the kind of error this page exists to catch: it named the wrong hypothesis for a citation to discharge.*

Provenance. This page was written after an audit found the same curvature *claim* consumed in three places under three different pairings of measure and operator, and after a proposed repair — citing the metric-measure result in place of the regular-stratum one — was found to change the object rather than fix the proof. A later pass found that the regular-stratum claim is not established by its own printed proof either, so the defect is now twofold: an invalid derivation in one paper, and a correct theorem in another about a different object. A third pass found the drift *on this page*: the row for `2602.0033` had recorded the printed measure against ν when the paper writes it against Vol_B, which is precisely the substitution (I2b) now forbids. It was introduced by writing the row from the intended repair rather than from the printed text. Hence the rule below, and hence (I2b) as a numbered step rather than a remark. The same pass then caught the correction overshooting: (I2b) had asserted the two measures *are* different, which needs w non-constant and is not proved here. Both directions are now stated at the strength they have. A fourth pass found that (I3) wrote its Hamiltonian as $\frac{g^2}{2} K_\nu + V$ — and V is already bound, in `2602.0035`, to the gauge-fixed $S_{\mathrm{YM}} - \log\det M$. A page reproduced verbatim inside that paper cannot borrow a symbol the host has spent. The first repair reached for a bare W, which that same paper binds to the Weyl group S_{N_c}: **no bare letter is safe in a page meant to be embedded, only a subscripted name**. The same pass found the collision that matters most, because it is in the *measure* column: both host papers write "the uniform measure ν" for the normalised $d\mathrm{Vol}_B$, which is what this page's own (I2b) forbids identifying with $\pi_\# \mathrm{Vol}_{\mathcal{A}}$. Hence the notation barrier above. A fifth pass found the row for `2602.0035` built from its intended repair rather than its printed text — the identical error this page had already corrected once, for `2602.0033` — and found the claim that that paper needs (I3) to be simply wrong: its v2 already uses a ground-state measure. It records a failure mode that is not covered by the usual distinction between a false statement and an invalid proof: **a correct theorem about the wrong object for the consuming dependency**. Reproduce it verbatim; do not paraphrase it, because paraphrase is how the columns drift.